Atlantic Ocean International Research Network Analysis

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Basic Statistics

- 46,559 different research articles published between 2000 and 2020
  - 17,213 (37%) articles were collabs between different countries
- 167 unique countries
  - 164 have collaborated with other countries
    - Excluding only Palau, Serbia and Montenegro, and Macedonia
- 4, 476 unique journals
- 1,180 unique fields of study

Journal Representation

Most Represented (Top 10)

journal	identifier	percent	article_count
GEOPHYS RES LETT	[551465400058, 462072800060, 445612500067, 306...	0.023239	1082
J GEOPHYS RES-OCEANS	[178923600002, 89921000001, 595724300018, 5300...	0.018729	872
AQUACULTURE	[566467100013, 566468000030, 566468000024, 566...	0.017268	804
J CLIMATE	[439919300006, 234801900012, 391856300022, 359...	0.016538	770
MAR ECOL PROG SER	[183325800008, 183325800009, 545928500016, 485...	0.015314	713
PLOS ONE	[355185600009, 406768200015, 355979500085, 596...	0.015013	699
ICES J MAR SCI	[412309400009, 308012000011, 305460200017, 305...	0.014047	654
J FISH BIOL	[254809200022, 255040800017, 578704700001, 573...	0.013231	616
ZOOTAXA	[354993000007, 278759300002, 262631000001, 364...	0.012565	585
DEEP-SEA RES PT II	[247585300008, 232212500009, 252385500018, 401...	0.011555	538

- Average article count is **10**
- Median article count is **2**

- Journal Representation is skewed

Subject Representation

Most Represented (Top 10)

subject_category	identifier	article_count
Oceanography	[449133700011, 322575600011, 247585300008, 242...	3171
Meteorology & Atmospheric Sciences	[520136600001, 459364100011, 439919300006, 399...	2588
Fisheries|Marine & Freshwater Biology	[254809200022, 319985300004, 255040800017, 227...	2461
Geosciences, Multidisciplinary	[551465400058, 462072800060, 445612500067, 367...	2297
Marine & Freshwater Biology	[406988800015, 306842400001, 238683900019, 180...	1770
Multidisciplinary Sciences	[573749000013, 526871700012, 443175500032, 355...	1696
Zoology	[354993000007, 306392900006, 278759300002, 262...	1539
Fisheries	[264087800027, 183604300002, 165750100008, 432...	1507
Geochemistry & Geophysics	[328593500011, 317822600014, 265594800020, 248...	1420
Marine & Freshwater Biology|Oceanography	[368444900004, 255604100008, 169251200004, 582...	960

- Average article count is **39**
- Median article count is **3**

- Subject Category is skewed

Recent Country Contributions (last 5 years)

High Recent Percentage (100%)

	Country	total articles	recent articles	recent %
122	CONGO DEMOCRATIC REPUBLIC	6	6	1.0
148	ANDORRA	1	1	1.0
151	RWANDA	1	1	1.0
147	ARMENIA	1	1	1.0
153	KUWAIT	1	1	1.0
146	BOSNIA & HERZEGOVINA	1	1	1.0
154	SOUTH SUDAN	1	1	1.0
145	MAURITIUS	2	2	1.0
150	BAHRAIN	1	1	1.0
130	BRUNEI	3	3	1.0
149	ZAMBIA	1	1	1.0
142	CENT AFR REPUBL	2	2	1.0
126	MALDIVES	5	5	1.0
152	TONGA	1	1	1.0
141	REPUBLIC OF GEORGIA	2	2	1.0
129	SYRIA	4	4	1.0
155	NEPAL	1	1	1.0
136	MOZAMBIQUE	3	3	1.0
156	SAMOA	1	1	1.0
144	LIBERIA	2	2	1.0
157	GUINEA	1	1	1.0

Low Recent Percentage (Bottom 10)

Country	total articles	recent articles	recent %
CONGO PEOPLES REP	4	1	0.250000
BULGARIA	20	5	0.250000
ANTIGUA & BARBUDA	19	5	0.263158
COTE IVOIRE	30	8	0.266667
UKRAINE	41	11	0.268293
OMAN	7	2	0.285714
GABON	10	3	0.300000
FINLAND	326	106	0.325153
BERMUDA	132	43	0.325758
SRI LANKA	6	2	0.333333

Country Collaborations

Collab Ratio by Total Articles

*Top 25 Contributors

Top 10 Independent workers

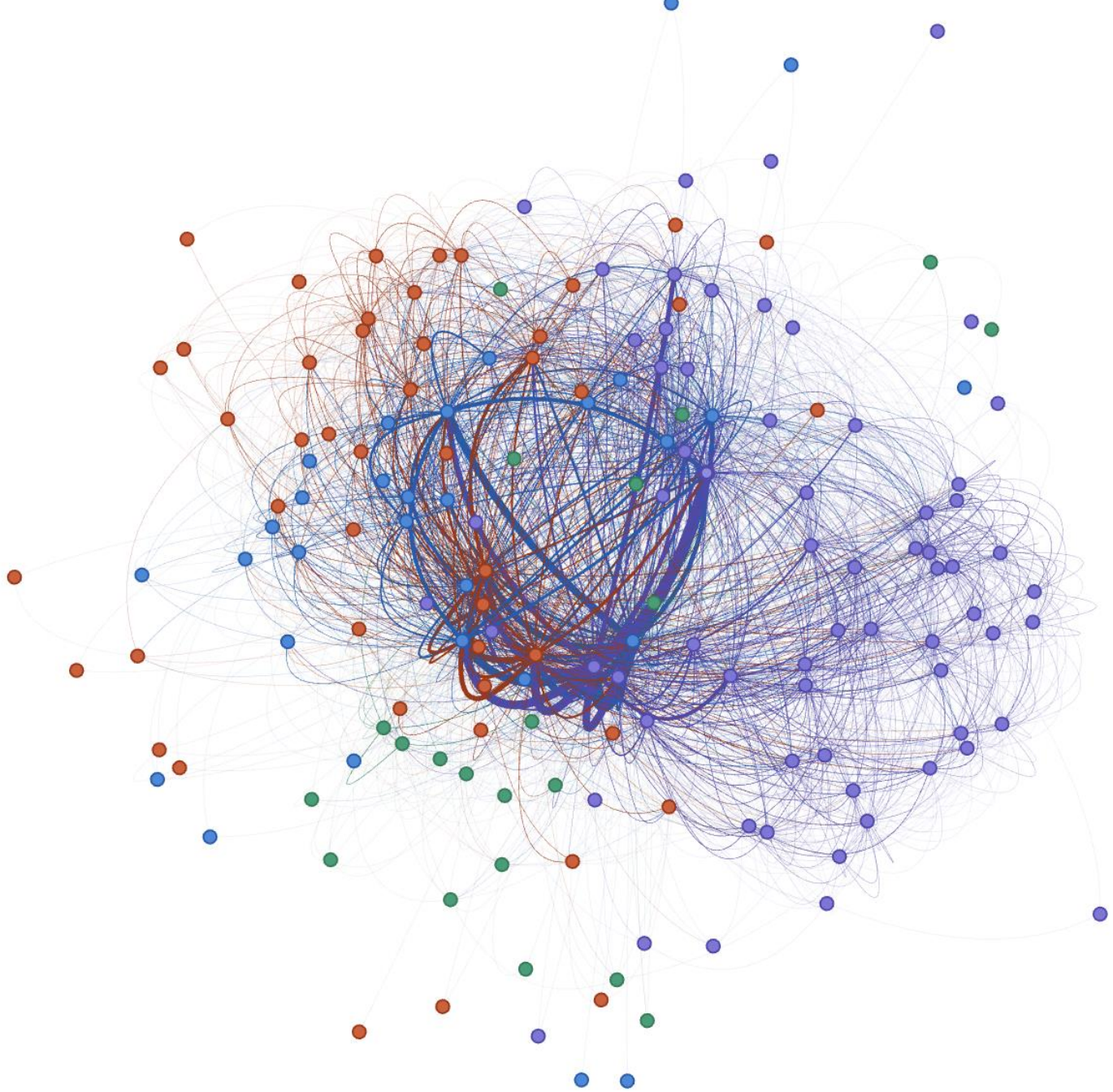
	Country	Count_contrib	Count_colab	Count_diff
3	BRAZIL	4629	1815	0.607907
29	TURKEY	315	131	0.584127
0	USA	15563	6742	0.566793
16	RUSSIA	892	388	0.565022
10	MEXICO	1412	698	0.505666
142	JORDAN	2	1	0.500000
143	UGANDA	2	1	0.500000
5	CANADA	4310	2246	0.478886
15	ARGENTINA	915	479	0.476503
17	COLOMBIA	849	449	0.471143
52	EGYPT	83	44	0.469880
30	JAMAICA	264	140	0.469697

Interconnectedness

Nodes:
All countries with at least
one collaboration

Edges:
Contribution to the same article
Weighted

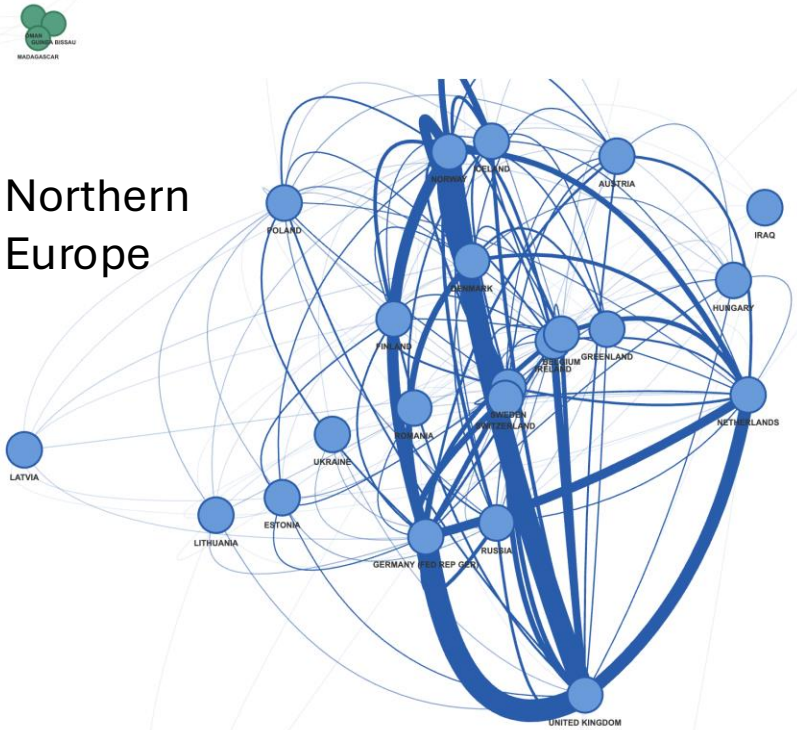
Clustering with:
louvaine_communities



Who works with who?

Southern Europe/ Middle East

Africa



Northern Europe

Americas

Ocean Future Objective

- Expand Research into underrepresented scientific fields like Biochemistry and Environmental Sciences
- Promote research in countries that have sped up their research in the past 5 years
- Bridge the gap between Americas (specifically Central and South America) and Europe.
- Brazil has particularly low collaboration ratio for academic output, good starting point
    - Followed by Russia, Mexico, Columbia
- Relatively low amount of academic output from Africa despite major Atlantic coastline.
    - South Africa is well connected, could act as bridge